QUALITY RISK ASSESSMENT + ENGINEERING STRATEGY

MapMyRun Field Quality Assessment

A balanced, evidence-bounded review of workout-record integrity, field reliability, user trust, and the engineering strategy for repeatable quality coverage.

PRIMARY ASSESSMENT	**Two Run records incorporated known vehicle travel and presented vehicle-scale pace as workout performance, including a 2:02 mile. This supports a credible UX and workout-integrity concern. Separately, the Gold Camp series produced three plausible saved walks under repeated GPS obstruction - a positive resilience result.**

ASSESSMENT TYPE	External black-box field assessment
PRIMARY RISK	User trust in saved workout records
CONFIDENCE	High in observed symptom; medium in product-quality interpretation; low in root-cause hypothesis
PROVISIONAL SEVERITY	Major / P2 candidate, subject to occurrence and downstream-impact data

Two independent test tracks. One balanced quality strategy.

External assessment | Evidence bounded | No internal access | Revised Aug. 24, 2026

01 / SCOPE AND METHOD

Assessment boundary and test strategy

This assessment evaluated the user-visible behavior of MapMyRun on an iPhone and Apple Watch. It used field notes, saved workout summaries, route maps, split tables, and analysis screens. It did not use source code, backend logs, raw GPS samples, internal requirements, production telemetry, or controlled location replay.

TWO INDEPENDENT QUALITY QUESTIONS

1) When a Run remains active during vehicle travel, does the saved record remain trustworthy, explainable, and recoverable? 2) Under repeated GPS obstruction and different phone/Watch acquisition paths, does recording degrade gracefully and still save a plausible workout?

Test matrix

ID	ACTIVITY / ACQUISITION	PHYSICAL CONDITION	PURPOSE
GC-01	Walk; iPhone-started; phone carried; Watch worn	Gold Camp Road; Tunnel #1 twice	Baseline phone acquisition under tunnel exposure
GC-02	Walk; Watch-started; phone left at Captain Jack's	Same practical course; Tunnel #1 twice	Compare supported Watch acquisition and later sync
GC-03	Walk; Watch-started; phone powered off at base	Same practical course; Tunnel #1 twice	Remove paired-phone availability during acquisition
SAT-WM	Run; recording remained active	Known driving plus pedestrian/errand behavior	Independent real-world occurrence
SUN-WM	Run; iPhone-started; Auto-Pause ON; Watch worn	Walk, drive, stop, shop, drive, walk	Targeted follow-up separating stop handling from semantic mismatch

Evidence-handling rules

- Field timestamps and saved-workout timestamps were preserved separately when they differed.
- A route following real roads was treated as evidence of geographically plausible location, not proof of raw-sample accuracy.
- Tunnel timing was aligned to analysis screens, but graph disturbances outside tunnel windows prevented simple causal attribution.
- Prior AI interpretations were treated as hypotheses and rechecked against the evidence ledger.
- No undocumented product requirement was invented to make the result look like a defect.

Important design caveats

The Gold Camp series was a controlled real-world comparison, not a laboratory A/B test. Durations, exact walking paths, satellite geometry, and environmental conditions were not identical. Saturday and Sunday were not identical reproductions; they were two distinct records showing the same broad vehicle-contamination signature.

TEST ENVIRONMENT

iPhone 17 Pro Max | iOS 26.6 | Apple Watch SE 3 GPS + Cellular, 44 mm | watchOS 26.6 (23U67) | MapMyRun 26.10.0

02 / EVIDENCE

What was observed - and what it supports

Controlled Gold Camp Road series

All three Walk workouts used Captain Jack's parking lot as base and traversed Tunnel #1 twice. Their aggregate pace, heart rate, cadence, and elevation gain were tightly grouped. Their instantaneous traces were noisy, including around some tunnel windows and elsewhere, but every saved workout remained physically plausible.

METRIC	GC-01	GC-02	GC-03
Acquisition	iPhone / carried	Watch / phone at base	Watch / phone off
Duration	18:25	20:15	21:34
Distance	0.91 mi	1.00 mi	1.07 mi
Avg pace	20:07/mi	20:18/mi	20:05/mi
Avg HR	125 bpm	124 bpm	124 bpm
Avg cadence	94 spm	92 spm	94 spm
Elevation gain	80 ft	75 ft	82 ft
Tunnel windows	12:08-10; 12:14-16	12:33-35; 12:41-42	1:02-03; 1:09-11

RED-TEAM CORRECTION

The 0.91-1.07 mi spread is not evidence of a distance defect by itself. Workout durations increased from 18:25 to 21:34 while average pace stayed near 20:10/mi; more recorded time at the same pace naturally produces more distance. The route was similar, but exact execution length was not controlled tightly enough to isolate distance accuracy.

Vehicle-contaminated Run records

This is a separate real-world test track. The Gold Camp series was not designed to explain or reproduce these records.

EVIDENCE	SATURDAY WALMART / ARTEM	SUNDAY WALMART
Ground truth	Known driving; note: 'Went to Walmart, then to Artem. Drove twice.'	Deliberate walk/drive/stop/shop/drive sequence
Saved summary	5.64 mi | 1:18:12 | 13:51/mi	5.25 mi | 29:21 | 5:35/mi
HR / cadence	85 bpm / 82 spm average	100 bpm / 94 spm average
Vehicle-scale pace	2:02/mi split (~29.5 mph); also 3:09 and 3:38	2:16, 2:20, 2:19/mi; max instantaneous 1:30/mi (~40 mph)
Route evidence	Map follows known roads traveled	Map follows recognizable roads; elevation tracks relocation
State handling	Not fully reconstructed	Auto-Pause ON; paused observed at 10:48 and 11:09

STRONGEST CROSS-METRIC EXAMPLE

Sunday mile 2 saved as 2:16/mi while the split table reported HR 84 bpm and cadence 98 spm. That combination is not a credible description of a human completing a 26.5 mph mile. It is strong anomaly evidence, though the provenance and intended semantics of every sensor value remain unverified.

03 / FINDINGS

Primary quality issue and severity rationale

Defect candidate

TITLE	Activity-inconsistent vehicle movement is incorporated into saved Run metrics without an observed data-quality signal
OBSERVED	Two separate Run records containing known automobile travel saved vehicle-scale pace/splits. Sunday included 2:16-2:20 miles and a 1:30/mi instantaneous maximum; Saturday included a 2:02 mile. The road-following routes and known travel make simple random GPS teleportation a weaker explanation.
EXPECTED	At the product-quality level, a saved workout should either remain representative of the selected activity or clearly communicate suspect segments and provide a practical recovery path. The exact product requirement and preferred behavior require internal product confirmation.
FREQUENCY	Broad behavioral signature observed in two of two available drive-containing Run records. This is not a controlled population rate or a claim of universal reproducibility.
CONFIDENCE	High that vehicle movement entered the saved records; medium that the lack of visible integrity signaling is a product-quality gap; low regarding the responsible component or algorithm.

Why this matters

Workout records feed progress history, pace trends, goals, coaching interpretation, social sharing, and potentially downstream platforms. A record can be technically complete yet semantically wrong. **A visibly broken workout is obvious. A polished-looking workout that is wrong is harder to detect and potentially more damaging to trust.** Because the values are precise and authoritative-looking, the user may not realize which portions are invalid or how much of the record remains trustworthy.

DIMENSION	ASSESSMENT
User impact	Materially misleading core workout metrics; legitimate and invalid segments become blended
Recoverability	Not established. Editing/trimming capability was not evaluated in the captured evidence
Scope	Observed twice on one user's device/account; fleet frequency unknown
Safety	No direct safety harm demonstrated
Data loss / crash	Neither observed
Downstream impact	Plausible but unverified; Apple Health/export propagation was not established
Provisional call	Major / P2 candidate; priority should be adjusted using production frequency, support signal, and propagation data

Why this is not an Auto-Pause defect

Auto-Pause is principally a stopped-state feature. On Sunday it was enabled, and the app was observed paused at 10:48 and 11:09. The saved duration was also much shorter than wall-clock execution. The stronger interpretation is that stationary handling may have operated while moving vehicle segments remained eligible. A speed/activity plausibility policy is a different product concern.

04 / HYPOTHESES AND LIMITATIONS

What each test track establishes - and leaves open

HYPOTHESIS	STATUS	RATIONALE
Gold Camp GPS-obstruction resilience	Positive result	Six tunnel passages produced some transient noise but three plausible saved walks
Acquisition-path resilience	Positive signal	Phone-started, Watch-started, and phone-off scenarios produced similar plausible aggregates
Active Run accepts vehicle movement	Observed twice	Known driving aligns with road routes and vehicle-scale splits
Raw GPS was random or grossly displaced	Less consistent	Routes followed recognizable roads, but raw samples were not inspected
Workout-integrity signaling is insufficient	Product / UX risk	Selected Run plus 26-40 mph derived pace saved without an observed integrity signal
Cross-sensor validation is absent or broken	Unknown	HR/cadence disagreement is visible; intended internal use of those streams is unknown
Watch, phone, or backend is responsible	Unknown	Component attribution requires logs, provenance, and raw time series
GC distance variation is a defect	Not established	Duration variation and uncontrolled endpoint precision are credible explanations

Alternative explanations considered

User error: The workout remained active during driving. That explains how contamination began, but does not eliminate the product question of how trustworthy records should behave after a realistic mode mistake.

Correct-by-design recording: The system may intentionally record every valid coordinate until the user stops it. If so, the issue may be a product/design gap rather than a violated implementation requirement.

Sensor artifacts: Cadence during driving may reflect motion artifacts, smoothing, or display semantics; it should not be treated as independent ground truth without raw samples.

GPS noise: Road-following geometry and repeated vehicle-scale splits make purely random drift less compelling, but only raw time series can distinguish accepted vehicle movement from reconstruction behavior.

Wrong activity selection: Both observed records were Run workouts. Selecting Run does not make 26-40 mph miles plausible human running.

Known evidence gaps

GAP	WHY IT MATTERS
No raw timestamped location/speed/accuracy series	Prevents segment-level attribution and algorithm analysis
No device firmware detail beyond recorded OS/build	Avoids inventing environment details not preserved in the evidence
No documented requirement for anomaly handling	Keeps the expected result at product-quality level
No edit/recovery assessment	Limits severity and user-impact conclusions
No downstream validation	Cannot claim Apple Health, exports, calories, or coaching were corrupted
Single user/account/device ecosystem	No population or platform generalization
11:02 Watch state unresolved	Kept out of the main defect rather than inferred
Post-tunnel Apple notifications unresolved	Platform observation; no MapMyRun causality claimed

05 / QUALITY OPERATING MODEL

Turn field evidence into a managed quality system

QUALITY-LEAD PRIORITY

First establish prevalence, user impact, recovery options, and downstream propagation. Assign product ownership for workout-integrity policy and engineering ownership for traceability. Then convert both the vehicle-transition risk and the Gold Camp resilience baseline into deterministic automation, release gates, and production monitoring.

1. Reproduce deterministically at the data layer

Capture or replay a known-good walking/running trace, then inject vehicle segments at controlled speeds, durations, accelerations, and transition points. Compare raw samples, accepted segments, aggregates, UI, sync, exports, and downstream consumers. Include poor-accuracy points, true GPS jumps, urban canyons, tunnels, and stale fixes so semantic mismatch is not confused with sensor corruption.

TEST DIMENSION	BOUNDARIES / QUESTIONS
Activity-specific speed	Walk/Run thresholds; short sprint vs cycling vs vehicle; duration and confidence
Transitions	Pedestrian -> vehicle -> stopped -> pedestrian; pause/resume; background/foreground
Sensor availability	GPS only; GPS + HR; GPS + cadence; missing/stale/conflicting streams
Acquisition path	iPhone, Watch, paired/absent phone, offline sync, later reconciliation
Recovery UX	Warn, segment, trim, relabel, confirm, preserve original, undo
False positives	Elite sprinting, downhill travel, treadmills, wheelchairs, skating, transit near route
Data propagation	Summary, history, goals, coaching, calories, Apple Health, exports, social

2. Define the product policy before encoding thresholds

A hard speed cutoff may destroy legitimate data and create accessibility problems. Product, data science, support, privacy, and engineering should agree on the desired response: automatic exclusion, confidence scoring, review prompt, activity relabeling, or post-workout repair. The best solution may combine duration, acceleration, GPS accuracy, route context, and sensor agreement rather than a single mph rule.

3. Instrument the decision

Measure the rate of implausible activity/speed combinations, user edits or deletions after such events, support contacts, device/app segmentation, and downstream sync. Log why segments were accepted, rejected, or flagged using privacy-appropriate telemetry. This turns severity and priority from opinion into an evidence-based product decision.

4. Add regression properties

PROPERTY	EXAMPLE ACCEPTANCE CRITERION
Internal consistency	Aggregate distance/time/pace reconcile to accepted segments within defined tolerance
Explainability	Flagged or excluded segments have a stable reason and user-visible recovery path
Preservation	No legitimate segment is silently deleted; original data remains recoverable per policy
Idempotent sync	Offline Watch reconciliation does not lose, duplicate, or reclassify the workout
Uncertainty	Low-confidence data is not presented with stronger certainty than the system possesses

06 / MANAGEMENT VIEW

Balanced conclusions across quality dimensions

DIMENSION	ASSESSMENT	MANAGEMENT RESPONSE
Data integrity	Vehicle travel entered two saved Run records and produced physically implausible performance	Treat as a credible risk; establish prevalence and downstream impact
UX / trust	A 2:02 mile was presented as authoritative workout data without an observed integrity signal	Define warning, repair, trim, relabel, and recovery policy
Reliability	All three Gold Camp walks saved successfully and remained plausible	Preserve as a positive resilience baseline
Accuracy	Transient traces were noisy, but aggregate pace, HR, cadence, and elevation were stable	Set tolerances and replay representative degraded traces
Observability	External evidence cannot identify sample provenance, acceptance decisions, or responsible component	Add privacy-aware decision telemetry and trace IDs
Testability	Field testing is useful but not deterministic or scalable	Create trace replay, synthetic transitions, contracts, and regression properties
Release risk	Frequency and propagation remain unknown	Use telemetry and support data to set severity, gates, and ownership

Secondary observations retained but not promoted

GC distance variance: interesting during exploration, but not defect evidence after accounting for duration differences.
Watch workout-ending discoverability: one user experienced difficulty finding End after Pause; useful usability signal, insufficient for a broad conclusion without task-based testing.
Post-tunnel Apple notifications: repeated timing observation, but content and mechanism were not established.
11:02 Watch display: showed SPM 0 and pace unavailable after the nominal Sunday workout end; exact session state remains unknown and is excluded from the primary finding.

FINAL ASSESSMENT

The vehicle-transition evidence supports a credible UX and workout-integrity concern: authoritative-looking Run records preserved known vehicle movement, including a 2:02 mile, without an observed integrity signal. Independently, the Gold Camp series showed respectable reliability and graceful degradation under repeated GPS obstruction. The cause of the contaminated records remains unresolved. The next step is a managed quality program: define product policy, establish prevalence, replay traces deterministically, automate cross-layer checks, and monitor production outcomes.

AI assistance disclosure

AI accelerated background research, test-design iteration, evidence normalization, challenge of competing interpretations, and document production. Michael Jensen executed the field tests, captured observations, reviewed the evidence, challenged incorrect interpretations, and retained ownership of the final conclusions.

Evidence provenance

Prepared from the Aug. 22-23, 2026 field notes and saved MapMyRun summary/analysis screenshots documented in the QA Career Comeback Plan and Observation Photo Analysis task handoff. This report intentionally preserves uncertainty where the available record is incomplete.

What I would change in the next field cycle

THE MOST IMPORTANT LESSON

The three Gold Camp paths were very similar, but not controlled tightly enough to attribute small distance differences to MapMyRun. That limits causal confidence; it does not invalidate the positive reliability result. Recognizing the limitation and redesigning the protocol is disciplined quality engineering.

LESSON	WHY IT MATTERED	NEXT PROTOCOL
Synchronize starts	Split capture competed with attention to other controls.	Synchronize clocks; start every trial at the top of a minute.
Fix the physical route	A small route change can explain a small distance difference.	Mark exact start, turnaround, and end points; follow the same line.
Predefine variables	Exploration can drift as observations accumulate in the field.	Write the hypothesis, controls, and acceptance criteria before departure.
Repeat conditions	One trial per path cannot separate condition effects from variation.	Run multiple trials per condition and repeat on another day when practical.
Separate logging	Recording split times increased cognitive load while walking.	Use a checklist, voice timestamps, lap marker, or second observer.
Capture deviations	A remembered difference becomes hard to quantify later.	Log every pause, route change, interaction, and anomaly immediately.
Preserve raw evidence	Saved summaries cannot reveal filtering or sample provenance.	Retain the highest-resolution tracks, screenshots, clocks, and conditions.

A stronger repeatable field protocol

Before testing, create a one-page run card with the exact route, device state, start rule, interaction sequence, observations to capture, and abort criteria. At the site, mark the control points, synchronize clocks, and execute the card without improvising. After each trial, record deviations before reviewing results. Only then compare conditions and decide whether the evidence supports a product conclusion, a new hypothesis, or another experiment.

WHY THIS IS A STRENGTH

Field science is iterative. The useful outcome is not pretending the first design was perfect; it is showing where inference became weaker, preserving the uncertainty, and improving the next experiment. That inspect-and-adapt loop is how I would lead product-quality work with an engineering team.